**Solution**

Consider the set of all **even integers** $\mathbb{Z}_e = \{2k \mid k \in \mathbb{Z}\} = \{..., -6, -4, -2, 0, 2, 4, 6, ...\}$.

Let us verify the properties satisfied by $+$ on $\mathbb{Z}_e$.

(i) The sum of any two even integers is also an even integer.

Because $x, y \in \mathbb{Z}_e \Rightarrow x = 2m$ and $y = 2n$, $m, n \in \mathbb{Z}$.

So $(x + y) = 2m + 2n = 2(m + n) \in \mathbb{Z}_e$. Hence $+$ is a binary operation on $\mathbb{Z}_e$.

(ii) $\forall x, y \in \mathbb{Z}_e, (x + y) = 2(m + n) = 2(n + m) = (2n + 2m) = (y + x)$.

So $+$ has commutative property.

(iii) Similarly it can be seen that $\forall x, y, z \in \mathbb{Z}_e$, $(x + y) + z = x + (y + z)$.

Hence the associative property is true.

(iv) Now take $x = 2k$, then $2k + e = e + 2k = 2k \Rightarrow e = 0$.

Thus $\forall x \in \mathbb{Z}_e$, $\exists 0 \in \mathbb{Z}_e \ni x + 0 = 0 + x = x$.

So, $0$ is the identity element.

(v) Taking $x = 2k$ and $x'$ as its inverse, we have $2k + x' = 0 = x' + 2k \Rightarrow x' = -2k$. *i.e.,* $x' = -x$.

Thus $\forall x \in \mathbb{Z}_e, \exists -x \in \mathbb{Z}_e \ni x + (-x) = (-x) + x = 0$

Hence $-x$ is the inverse of $x \in \mathbb{Z}_e$.

**Example 12.5**

Verify the (i) closure property, (ii) commutative property, (iii) associative property (iv) existence of identity and (v) existence of inverse for the arithmetic operation $+$ on $\mathbb{Z}_o$ = the set of all odd integers.

**Solution**

Consider the set $\mathbb{Z}_o$ of all **odd integers** $\mathbb{Z}_o = \{2k + 1 : k \in \mathbb{Z}\} = \{..., -5, -3, -1, 1, 3, 5, ...\}$. $+$ is **not a binary operation** on $\mathbb{Z}_o$ because when $x = 2m + 1, y = 2n + 1, x + y = 2(m + n) + 2$ is even for all *m* and *n*. For instance, consider the two odd numbers $3, 7 \in \mathbb{Z}_o$. Their sum $3 + 7 = 10$ is an even number. In general, if $x, y \in \mathbb{Z}_0$, then $(x + y) \notin \mathbb{Z}_0$. Other properties need not be checked as it is not a binary operation.

**Example 12.6**

Verify (i) closure property (ii) commutative property, and (iii) associative property of the following operation on the given set.

$(a * b) = a^b; \forall a, b \in \mathbb{N}$ (exponentiation property)

**Solution**

(i) It is true that $a * b = a^b \in \mathbb{N}; \forall a, b \in \mathbb{N}$. So $*$ is a **binary operation** on $\mathbb{N}$.

(ii) $a * b = a^b$ and $b * a = b^a$. Put, $a = 2$ and $b = 3$. Then $a * b = 2^3 = 8$ but $b * a = 3^2 = 9$

So $a * b$ need not be equal to $b * a$. Hence $*$ **does not have commutative property**.

(iii) Next consider $a * (b * c) = a * (b^c) = a^{(b^c)}$. Take $a = 2$, $b = 3$ and $c = 4$.

Then $a * (b * c) = 2 * (3 * 4) = 2^{3^4} = 2^{81}$

But $(a * b) * c = (a^b) * c = (a^b)^c = a^{(bc)} = a^{bc} = 2^{12}$

Hence $a * (b * c) \neq (a * b) * c$. So $*$ **does not have associative property** on $\mathbb{N}$.

**Note:** This binary operation has no identity and no inverse. (Justify).



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